

Tn-seq: high-throughput parallel sequencing for fitness and genetic interaction studies in microorganisms

Tim van Opijnen, Kip L Bodi & Andrew Camilli

Biological pathways are structured in complex networks of interacting genes. Solving the architecture of such networks may provide valuable information, such as how microorganisms cause disease. Here we present a method (Tn-seq) for accurately determining quantitative genetic interactions on a genome-wide scale in microorganisms. Tn-seq is based on the assembly of a saturated Mariner transposon insertion library. After library selection, changes in frequency of each insertion mutant are determined by sequencing the flanking regions *en masse*. These changes are used to calculate each mutant's fitness. Using this approach, we determined fitness for each gene of *Streptococcus pneumoniae*, a causative agent of pneumonia and meningitis. A genome-wide screen for genetic interactions of five query genes identified both alleviating and aggravating interactions that could be divided into seven distinct categories. Owing to the wide activity of the Mariner transposon, Tn-seq has the potential to contribute to the exploration of complex pathways across many different species.

Genes (and their protein products) are organized in complex networks. Genetic interaction analysis on a genome-wide scale has the potential to connect genes and to increase understanding of such networks. Interactions between genes have been explored in *Saccharomyces cerevisiae* by systematically knocking out all possible pairwise combinations of genes, leading to identification of networks in important pathways such as DNA integrity¹ and phosphorylation². Such comprehensive genetic interaction screens are rare in microorganisms owing to a lack of genome-wide tools with wide applicability. Thus the development of a high-throughput tool for gene disruption and interaction discovery in microorganisms has great potential for improving the lagging functional annotation of microbial genomes and for resolving complex biological pathways.

Three approaches have been developed to identify genetic interactions in *S. cerevisiae*^{3–6}. Each makes use of an arrayed collection of single-gene deletion strains and robotic automation to construct double mutants by mating. Meiotic assortment is

not applicable to bacteria, but analogous methods have been developed for *Escherichia coli*: genetic interaction analysis technology for *E. coli* (GIANT *coli*) and *E. coli* synthetic genetic array (eSGA)^{7,8}. These methods use conjugation of a genome-wide arrayed collection of single gene mutants to construct all double mutants. However, their applicability to a wide range of microorganisms is problematic, as there are arrayed collections of single gene knockouts for only a handful of species.

A more generally applicable method to identify genetic interactions in bacteria is transposon site hybridization (TraSH)⁹ as well as related transposon-based genetic footprinting methods¹⁰. TraSH is based on the construction of a transposon library, the generation of probes from outward-oriented T7 promoters located in the transposon ends and the subsequent detection of the probes by microarray. However, the accuracy of this and the related methods is somewhat limited because the precise location of an insertion cannot be directly determined, fitness cannot be determined for each insertion but is depicted as a relative fluorescence intensity ratio, and the need for a microarray restricts their use to strains for which microarrays have been developed.

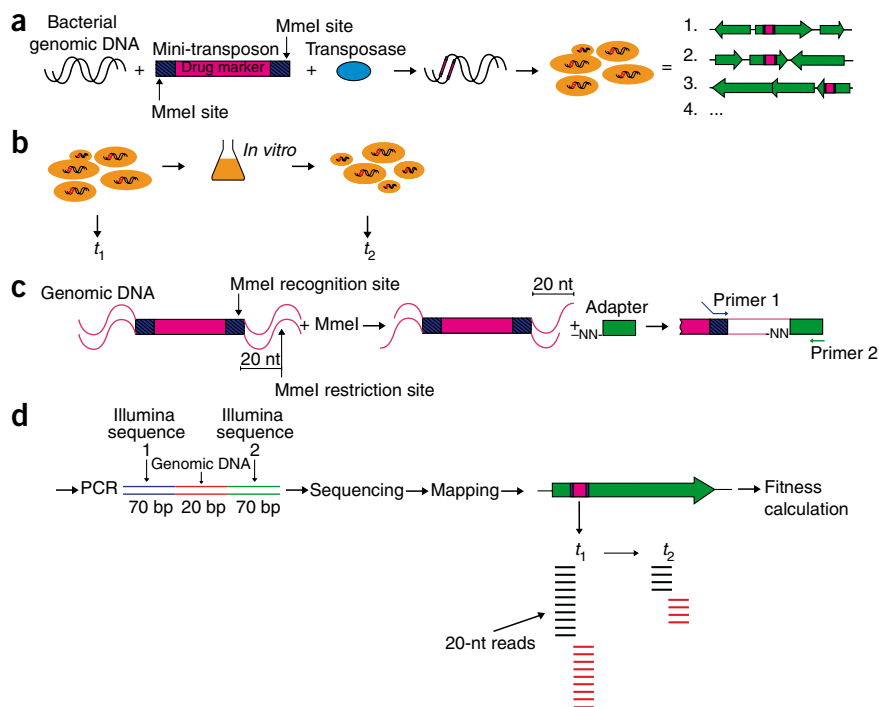
Here we present Tn-seq, a robust and sensitive method for the discovery of quantitative genetic interactions in microorganisms through massively parallel sequencing. The approach does not depend on a preexisting genomic microarray or an array of gene knockout strains but is instead based on the assembly of a saturated transposon insertion library. After growth of the library under a test condition, the change in frequency of each insertion mutant is determined by sequencing the transposon flanking regions *en masse*. The change in frequency reflects the effect of the insertion on fitness. Fitness of every insertion in a genome can be determined in this way and is a quantitative measure of the growth rate. We applied Tn-seq to the gram-positive bacterium *S. pneumoniae*, determining fitness for each gene and identifying those likely to be essential for basal growth. A genome-wide screen for genetic interactions of five query genes involved in transcriptional regulation and carbohydrate transport identified 97 high-confidence interactions.

Howard Hughes Medical Institute, and Department of Molecular Biology and Microbiology, Tufts University School of Medicine, Boston, Massachusetts, USA. Correspondence should be addressed to A.C. (andrew.camilli@tufts.edu).

RECEIVED 24 JULY; ACCEPTED 31 AUGUST; PUBLISHED ONLINE 20 SEPTEMBER 2009; DOI:10.1038/NMETH.1377

Figure 1 | Schematic depiction of Tn-seq.

(a) A gene disruption library is constructed by first transposing the mini-transposon *magellan6*, which contains an *Mme* restriction site within each inverted repeat, into bacterial genomic DNA *in vitro* and then transforming a bacterial population with the transposed DNA. The result is a bacterial pool in which each bacterium contains a single transposon insertion in its genome. (b) DNA is isolated from a portion of the bacterial pool (t_1) and another portion is used to seed a culture on which selection is performed, then DNA is isolated again from recovered bacteria (t_2). (c) DNA from both time points is digested with *Mme*I. (d) A PCR amplification was performed to obtain a 160-bp sequence with 20 bp of bacterial-specific DNA flanked by Illumina-specific sequences, which enable sequencing. After sequencing, different samples are identified based on barcode sequence, and the 20-bp reads are mapped to the genome and are counted for each insertion, thus allowing fitness to be calculated.



RESULTS

Library selection, sequencing and fitness calculations

We established a genome-wide disruption library in *S. pneumoniae* with the Mariner Himar1 mini-transposon derivative *magellan6*. *Magellan6* inserts randomly in the genome, requiring only a TA dinucleotide at the insertion site¹¹. After transposition *in vitro*, we transformed naturally competent *S. pneumoniae* with the transposed DNA, resulting in a library in which each bacterium carried a single transposon insertion (Fig. 1). We grew the library in standard broth medium, and during the growth, detrimental insertions decreased in frequency whereas advantageous ones

increased. We determined the change in frequency over time for each mutant by *en masse* sequencing of the regions flanking the transposon insertions, such that a change in fitness translated into a change in the number of reads. We used this change to calculate fitness for each insertion mutant (Online Methods).

Single gene fitness

To screen for genetic interactions it is necessary to measure the fitness effects of double gene mutations. First we need to measure fitness for each single gene disruption. For this, we constructed 6 libraries of ~25,000 insertions each in the wild-type *S. pneumoniae* TIGR4. We grew the libraries in standard broth for ~7 generations after which we isolated DNA and sequenced the transposon junctions. There was no insertional bias between coding and noncoding genomic regions, and there were 36 regions that had a $\geq 2\times$ higher or lower number of insertions (Supplementary Table 1). Figure 2a illustrates how different transposon insertions distributed over a specific genomic region affect fitness. By averaging over all insertions in a specific gene we obtained a single fitness value for each gene (Supplementary Table 2). We divided the genes into four categories; neutral (fitness = 0.96–1.04), advantageous (fitness > 1.04), disadvantageous (fitness < 0.96) and possibly essential (fitness = 0). Genes in the latter category correspond to those in which transposon insertions were absent in the sequenced library. Mutations in these genes either prevented or substantially slowed bacterial replication during outgrowth of the transposon insertion strains. Categorization was based on a one sample *t*-test with Bonferroni correction for multiple testing (Fig. 2b and Online Methods). We confirmed 68% of genes that were previously suggested to be essential in another serotype strain, R6 (Supplementary Table 2). The 32% of unconfirmed genes may be due to strain differences.

We used the gene set enrichment analysis (GSEA)¹² program to test for the enrichment of genes with a common biological function and a similar fitness effect. Eleven gene sets were

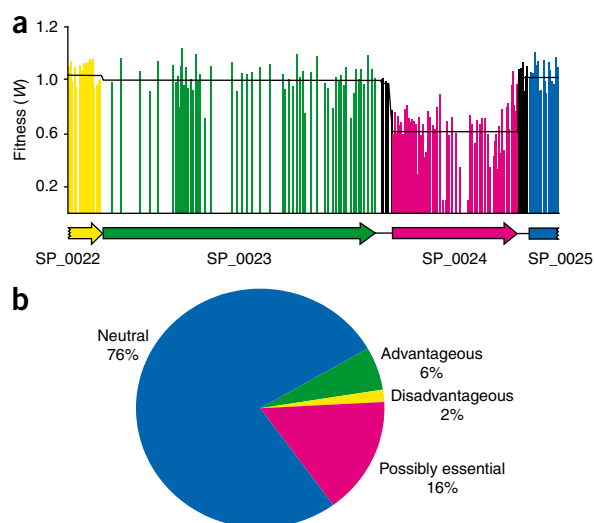
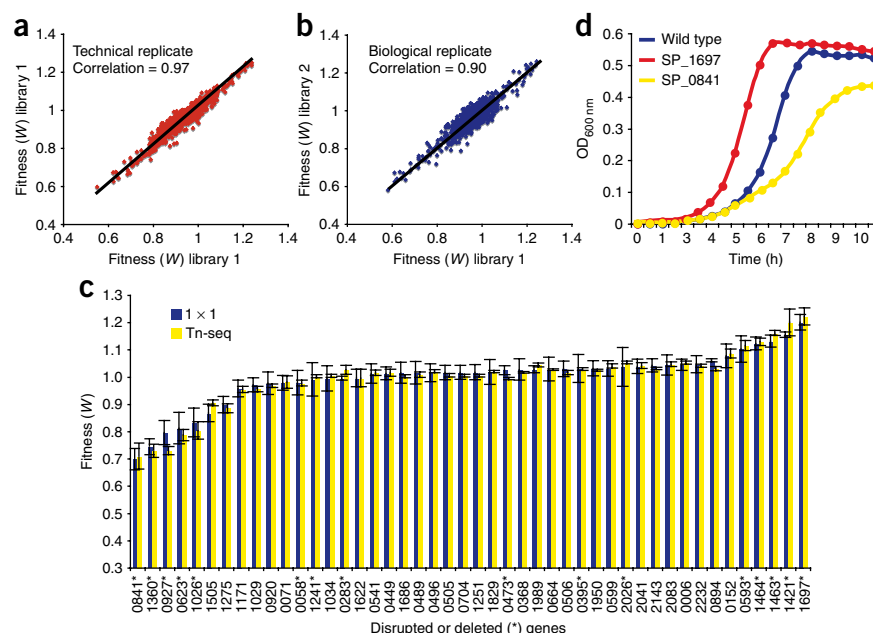


Figure 2 | Fitness determination and classification. (a) Independent insertions and their fitness effects in the genomic region from gene SP_0022 to SP_0025 are plotted. The black jagged line depicts the average fitness for each gene. (b) Categorization of every gene's fitness in the *S. pneumoniae* genome into four categories.

Figure 3 | Validation of Tn-seq fitness measurements. (a) A technical replicate of the same library comparing the effect of sample preparation and sequencing. (b) A biological replicate comparing two independently generated, selected upon and sequenced libraries. (c) Comparison of fitness ($\pm$ s.e.m.) obtained with the classical 1×1 competition method ($n = 5$) and Tn-seq (see **Supplementary Table 2** for n values), for 30 random transposon insertions and 16 marked gene deletions (*). Numbers under the graph refer to *Streptococcus pneumoniae* TIGR4 gene numbers (SP_number). (d) Representative growth curves for the wild-type strain, a strain with an advantageous deletion (SP_1697, fitness = 1.22) and a strain with a disadvantageous deletion (SP_0841, fitness = 0.71).



negatively enriched, suggesting that they are advantageous for growth in this environment and two were positively enriched (FDR < 25%, nominal $P < 0.01$), suggesting that they are deleterious for growth in this environment (**Supplementary Table 3**).

Validation of the method

To validate the reproducibility of Tn-seq, we performed technical and biological replicates. Technical replicates compared the same libraries that we split after DNA isolation. Reproducibility among technical replicates was very high (**Fig. 3a**). Biological replicates, which we performed for all (six) experiments in this study, compared two or more independently established and selected-on libraries. Reproducibility between biological replicates was also high (Pearson correlation coefficients were 0.70–0.90; a representative experiment is shown in **Fig. 3b**). These data show that Tn-seq is highly reproducible both within and between independent experiments.

We did not identify genes with fitness less than 0.53 but greater than 0 (**Fig. 3b** and **Supplementary Table 2**). As the generation time of such mutants was at least twice that of wild type, it is likely that such strains were rapidly outcompeted during construction and initial expansion of the library to the point that they fall below the threshold of detection by sequencing. Hence, the ‘possibly essential’ genes listed in **Supplementary Table 2** include both essential genes and genes that are highly deleterious when disrupted.

Table 1 | Doubling time and fitness values for strains

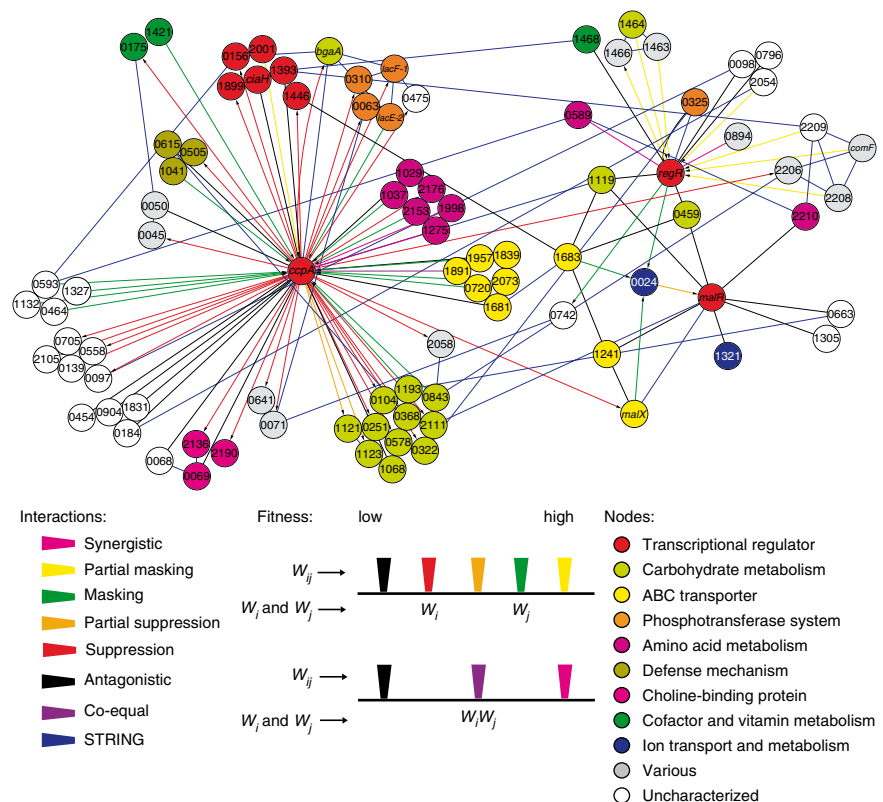
Strain	Fitness $\pm$ s.e.m. (n value)	Doubling time $\pm$ s.e.m. (min) ($n=4$)
Wild type	1	38.3 $\pm$ 1.0
SP_1697	1.22 $\pm$ 0.031 (11)	31.2 $\pm$ 1.6
SP_1464	1.13 $\pm$ 0.013 (37)	33.0 $\pm$ 0.9
SP_1463	1.16 $\pm$ 0.009 (69)	34.1 $\pm$ 0.8
SP_1421	1.20 $\pm$ 0.049 (10)	31.6 $\pm$ 0.8
SP_0593	1.11 $\pm$ 0.019 (62)	33.6 $\pm$ 0.8
SP_1360	0.73 $\pm$ 0.024 (12)	58.8 $\pm$ 3.0
SP_1026	0.80 $\pm$ 0.032 (21)	49.3 $\pm$ 1.4
SP_0927	0.73 $\pm$ 0.015 (42)	55.6 $\pm$ 3.5
SP_0841	0.71 $\pm$ 0.048 (6)	57.0 $\pm$ 2.8
SP_0623	0.79 $\pm$ 0.020 (36)	48.2 $\pm$ 1.6

It is possible that fitness determined using Tn-seq differs from fitness determined using the classical 1×1 competition. For instance, frequency-dependent selection, an evolutionary process in which fitness of a genotype is dependent on its frequency relative to other genotypes in the population, may be important. To address this possibility we measured fitness using 1×1 competition for 30 mutants that we randomly picked from the library. We observed no significant differences in fitness between data obtained using the two methods in a Student’s t -test (all $P \gg 0.05$) (**Fig. 3c**).

Depending on the site of insertion in a gene, a transposon can affect fitness in a position-dependent manner¹³. For example, insertions close to the 5′ or 3′ end may not disrupt the gene because of alternate start sites or truncated but still functional gene products, respectively, and could give a false indication of fitness. However, in Tn-seq, fitness was determined by averaging over multiple independent insertions in the same gene (**Fig. 2a**), minimizing the influence that positional effects have on fitness determination. To test for positional effects we re-analyzed the complete dataset by removing all insertions in the first or last 10% of each gene. Removing insertions in the first 10% of each gene did not change fitness and removing insertions in the last 10% changed fitness marginally for only four genes (**Supplementary Table 2**). This confirmed that positional effects were rare.

Transposon insertions within an operon may exert a polar effect on downstream genes leading to incorrect fitness assignments. Typically, such polar effects result from decreased expression of downstream gene(s). However, *magellan6* lacks transcriptional terminators therefore allowing for read-through transcription and thus such polar effects should be minimized. Although it was not possible to screen for the presence of polar effects, we could screen for their absence. The absence of polarity is most apparent when a downstream gene in an operon has a severe fitness effect or is essential whereas the upstream gene has no fitness effect. We screened the genome for operons with an essential downstream gene and identified 37 such instances. Of those 37 operons, none contained an insertion in an upstream gene with a (severe) fitness

Figure 4 | Genetic interaction network of query genes *ccpA* (SP_1999), *malR* (SP_2112), *malX* (SP_2108), *regR* (SP_0330) and hypothetical ABC transporter gene SP_1683. Interactions between two genes (*i* and *j*) are depicted as a line (edge) between two genes (nodes) and were determined by Tn-seq and supplemented with interaction data from the STRING¹⁹ database (confidence > 0.5; only edges were added and no nodes). Genetic interactions were divided into seven categories depicted under the network. W_i , fitness of mutant in gene *i*; W_j , fitness of mutant in gene *j*; W_{ij} , fitness of double mutant. In the top section, genes *i* and *j* confer different fitness; in the bottom section, genes *i* and *j* have an equal fitness effect. This scheme results in seven color coded genetic interaction categories: synergistic ($W_i < W_j < W_{ij}$), partial masking ($W_i < W_j < W_{ij}$), masking ($W_i < W_j = W_{ij}$), partial suppression ($W_i < W_{ij} < W_j$), suppression ($W_i = W_{ij} < W_j$), antagonistic ($W_j < W_i \leq W_{ij}$) and co-equal ($W_i = W_j = W_{ij}$). Arrows depict the hypothesized direction of the interaction. Nodes are color coded according to their involvement in a biological process or molecular function. Numbers in nodes refer to *Streptococcus pneumoniae* TIGR4 gene numbers (SP_number). Additional genes in nodes are *bgaA* (SP_0648), *ciaH* (SP_0799), *comF* (SP_2207), *lacE-2* (SP_1185) and *lacF-1* (SP_0476).



defect, suggesting that polar effects are not a major confounder in our data (Supplementary Table 2).

To confirm that Tn-seq accurately predicted fitness and also that a complete removal of a gene gave the same effect as averaging over multiple independent transposon insertions into the gene, we replaced the entire coding sequence of 16 genes with a drug marker. Of these, six had a Tn-seq-determined fitness of 0.95–1.05, five had a disadvantageous fitness of ≤ 0.8 and five had an advantageous fitness of ≥ 1.11 . We compared fitness between Tn-seq and the conventional 1×1 competition method and found no significant differences between the two methods in a Student's *t*-test (all *P* values $\gg 0.05$; Fig. 3c).

Because the expansion factor of the population over time was incorporated into the calculation of fitness, the resulting fitness value represented the change in frequency of a mutant in a population over a single generation. Thus fitness should correlate with the growth rate of the mutant strain. For instance, a 50% decrease in fitness ($W = 0.5$) should result in a twofold increase in doubling time. Confirming this, we determined the doubling times for 10 deletion mutants with reduced fitness, and these values directly correlated with fitness (Table 1 and Fig. 3d). For instance, given the wild-type strain doubling time of 38 min, a strain with a deletion of gene SP_1697 should have a doubling time of ~ 31 min ($(1/1.22) \times 38$) and we measured 31 min, and deletion of SP_0841 should yield a strain with a doubling time of ~ 54 min ($(1/0.71) \times 38$) and we measured 57 min.

Screening for genetic interactions

To assess the utility of Tn-seq in mapping genetic interactions, we performed a genome-wide screen on five 'query' genes. Three

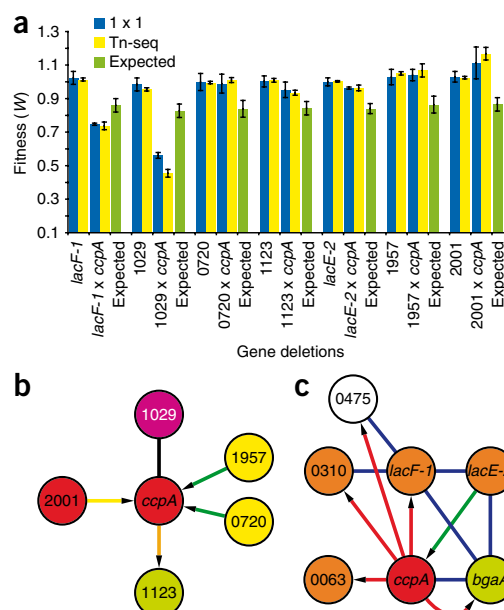
of these were transcriptional regulators (*ccpA*, *regR* and *malR*) and two are ABC transporters involved in carbohydrate uptake (SP_1683 and *malX*). The former are all members of the LacI/GalR family of metabolic regulators.

In *Bacillus subtilis*, catabolite control protein A (CcpA) is a master regulator of genes involved in complex carbohydrate metabolism and controls gene expression by binding to catabolite response DNA elements (*cre*). Depending on where a *cre* site is located, either upstream or downstream of the transcription start site, CcpA can, respectively, activate or repress transcription¹⁴. In *S. pneumoniae* inactivation of *ccpA* results in decreased *in vitro* fitness (0.84 ± 0.02 ; s.e.m.; $n = 24$) and decreased virulence in mouse models of colonization and infection^{15,16}.

Loss of *regR* has a less dramatic phenotype than *ccpA*: although it has no *in vitro* growth defect, it is involved in competence induction and virulence¹⁷. The third regulator MalR has so far only been implicated in the regulation of maltose uptake and metabolism¹⁸. The hierarchy among these three regulators led us to predict that we would find the most genetic interactions for *ccpA*, followed by *regR* and then *malR*. We also hypothesized that the genes encoding dedicated ABC transporters (SP_1683 and *malX*) would have still fewer genetic interactions. In addition, as the *malR* gene product regulates *malX* we expected to find overlap in the interactions of these two genes.

We replaced each of the five query genes with a drug marker and measured the fitness of each of the mutant strains in 1×1 competition experiments and confirmed these results by Tn-seq. Then we established three independent transposon libraries consisting of 10,000–25,000 insertions in the background of each query gene knockout. We used Tn-seq to determine fitness of all genes in each library. We scored alleviating and aggravating genetic interactions

Figure 5 | Validation of genetic interactions. **(a)** Fitness ($\pm$ s.e.m.) of seven genes either knocked out singly or in combination with *ccpA* are compared between the 1×1 method ($n = 5$) and Tn-seq (see **Supplementary Table 4** for n values), and the expected multiplicative fitness for each double gene knockout is given (expected fitness was determined by multiplying *ccpA* fitness (0.84) by fitness measured for the other gene). Numbers refer to *Streptococcus pneumoniae* TIGR4 gene numbers (SP_number). **(b)** A subnetwork (from the genetic interaction network shown in **Figure 4**) showing five of the validated genetic interactions and their specific interactions. **(c)** A subnetwork showing four phosphotransferase system genes including two of the validated genetic interactions (*lacF-1* (SP_0476) and *lacE-2* (SP_1185)), β -galactosidase (*bgaA* (SP_0648)) and a gene with unknown function (SP_0475). Interaction colors, node colors and numbers are as in **Figure 4**.



as a deviation from the multiplicative model and divided all interactions into seven categories based on a Student's *t*-test with Bonferroni correction for multiple testing (**Fig. 4**, **Supplementary Table 4** and Online Methods). We ordered the relationships among genes in a single network (**Fig. 4**) and supplemented these data with 41 additional interactions from the STRING¹⁹ database of known and predicted protein-protein interactions.

We scored the most interactions for *ccpA* (64) followed by *regR* (17) and *malR* (8), whereas the two transporters, SP_1683 and *malX*, had 5 and 3 interactions, respectively (**Fig. 4** and **Supplementary Tables 4,5**), the latter having two of its three interactions in common with *malR*. We also observed considerable overlap in interactions between SP_1683 and *malX* (2 of 3), between SP_1683 and *malR* (4 of 5) and between SP_1683 and *regR* (3 of 5). The overlapping profiles suggest that all four query genes are involved in overlapping metabolic pathways, which we expected for *malR* and *malX* and possibly SP_1683 (as, like *malX*, it is also involved in carbohydrate uptake) but not for *regR*.

The genetic interaction screen confirmed two expected interactions for *regR*; the STRING database-listed interaction with the mannose phosphotransferase system (SP_0325) and, by means of SP_2207 (*comF*) and SP_2208, the connection between *regR* and competence. Other *regR*-interacting genes suggest its involvement in various metabolic pathways as well as in other processes.

Of the five query genes, *ccpA* had the most genetic interactions detected, consistent with its role as a regulator of many different processes in *S. pneumoniae*. To determine whether there was a link between genes interacting with *ccpA* and the presence of a *cre* sequence, we screened the *S. pneumoniae* TIGR4 sequence for matches to the *B. subtilis* consensus *cre* sequence WTGNAANCGNWWNCA¹⁴, where W is A or T and N is A, C, G or T. We found 275 out of 2,087 genes (13%) with putative *cre* sequences (**Supplementary Table 2**). Out of the 90 genes present in the network, 27 have *cre* sequences, of which 24 interacted with *ccpA* (**Supplementary Table 4**). Thus *ccpA*, with 38% (24/64) of its interaction partners carrying a *cre* sequence, interacts with three times more *cre* sequence-containing genes than would be expected from chance alone.

Half of the genes (32/64) that interacted with *ccpA* are involved in carbohydrate uptake and metabolism. Other large categories include uncharacterized genes (13/64) and transcriptional regulators (6/64). Three of the latter are part of the two-component systems TCS07 (SP_0156), TCS11 (SP_2001) and TCS05 (SP_0799, *ciaH*). Two-component systems mediate communication between the environment and the cell interior. There is little known about

TCS07 and TCS11 but TCS05 (*CiaR/H*) has been found to regulate transcription of maltose utilization genes²⁰. *CiaR/H* is also needed for *S. pneumoniae* to colonize the nasopharynx and is important for lung and systemic infections in mice^{21,22}. We found an antagonistic interaction between *ciaH* and *ccpA* (fitness 0.46 ± 0.04 ; s.e.m.; $n = 12$) and suppressive interactions between *ccpA* and the maltose utilization genes *malX* (fitness 0.78 ± 0.02 ; s.e.m.; $n = 37$) and *malA* (SP_2111) (fitness 0.71 ± 0.03 ; s.e.m.; $n = 14$). We interpret these interactions to indicate that *ciaR/H* and *ccpA* have overlapping genetic interaction profiles. Indeed, *ciaR* has a predicted *cre* sequence, suggesting it is regulated by CcpA.

Validation of genetic interactions

To validate the genetic interactions identified by Tn-seq, we deleted seven genes by replacing them with a drug marker both in the wild-type and *ccpA* backgrounds. We confirmed fitness values for all 14 mutant strains by 1×1 competitions based on a Student's *t*-test with Bonferroni correction for multiple testing (**Fig. 5a**). Specifically, we confirmed alleviating interactions for SP_2001 (TCS05), ABC transporters SP_1957 and SP_0720 and metabolism gene SP_1123, and we confirmed an antagonistic interaction for amino acid metabolism gene SP_1029 (**Fig. 5b**). Additionally, we confirmed two interactions between *ccpA* and the phosphotransferase system genes SP_0476 (*lacF-1*) and SP_1185 (*lacE-2*), which are part of a small subnetwork including four phosphotransferase system genes, the hydrolase *bgaA* (SP_0648) and an uncharacterized gene SP_0475 (**Fig. 5c**).

DISCUSSION

Owing to our method of library construction we were most successful in identifying nonessential genes with fitness above 0.53 and possibly essential genes with fitness equal to zero. Some of the genes listed as possibly essential may have a fitness between 0.53 and 0. Adjustments in library assembly should increase the dynamic range of the method (Online Methods).

We recorded 97 high-confidence genetic interactions for five query genes in *S. pneumoniae*, with *ccpA* emerging as a 'hub' gene. Similar to its role in *B. subtilis*, CcpA thus appears to be a master

regulator in *S. pneumoniae*. Because of the sensitivity of Tn-seq we identified different types of both aggravating and alleviating interactions. Specific genetic interactions can reveal direction of information flow in molecular pathways^{23,24}. Here it is tempting to hypothesize that genes with an alleviating *ccpA* interaction represent deleterious genes that are repressed by CcpA in the growth condition we tested. In this scenario, knocking out *ccpA* results in derepression of the deleterious gene and a reduction in fitness. Subsequently knocking out the derepressed gene restores fitness.

We identified a subnetwork containing interactions between *ccpA*, four phosphotransferase system genes, the β -galactosidase *bgaA* and the uncharacterized gene SP_0475. In other bacterial species, cytoplasmic β -galactosidases are important in lactose metabolism. However, in *S. pneumoniae* BgaA is localized at the cell surface, and has so far only been associated with virulence²⁵. From both the Tn-seq data and its interaction with *ccpA*, we determined that *bgaA* is important in basal growth, and through its association in the subnetwork with two lactose phosphotransferase system genes, that it may be involved in lactose metabolism. In addition, the occurrence of the uncharacterized gene SP_0475 in the same subnetwork suggests that it too may be involved in phosphotransferase system-regulated carbohydrate uptake and metabolism. We expect that expanding the number of query genes and environmental conditions will contribute substantially to the understanding of the general architecture of *S. pneumoniae* biological networks and will be very helpful in annotating the 30% of unknown genes.

The Tn-seq data presented here illustrate the potential of the developed method; each gene's fitness can be rapidly determined in a specific environment without the need to first construct a full genome knockout array. As *S. pneumoniae* consists of >90 different serotypes and a gene's specific role may be strain-dependent this is an important advantage. Moreover, Tn-seq can be easily applied to different environments (for example, different growth media, chemical or physical stresses, or infection models). Finally, the Himar1 Mariner transposon can be used in many other microorganisms including *E. coli*²⁶, *Staphylococcus aureus*²⁷, *Haemophilus influenzae*²⁸ and *Mycobacterium tuberculosis*²⁹. We thus expect that our method will open up the possibility to screen for genetic interactions across many different environments, strains and species.

METHODS

Methods and any associated references are available in the online version of the paper at <http://www.nature.com/naturemethods/>.

Accession codes. NCBI Short Read Archive: SRA009392.

Note: Supplementary information is available on the Nature Methods website.

ACKNOWLEDGMENTS

We thank L. Sonenshein, R. Isberg, R. Iyer and N. Vastenhouw for both discussions and comments on the manuscript, members of the Tufts University core facility for Illumina sequencing and members of the Camilli lab, particularly D. Lazinski, and A. Boorsma for helpful discussions. This work was supported by a grant from the Netherlands Organization for Scientific Research (NWO-Rubicon) to T.v.O. and by the Howard Hughes Medical Institute.

AUTHOR CONTRIBUTIONS

T.v.O. and A.C. designed research, analyzed the data and wrote the paper. T.v.O. performed research and K.L.B. contributed to data analysis.

Published online at <http://www.nature.com/naturemethods/>.

Reprints and permissions information is available online at <http://npg.nature.com/reprintsandpermissions/>.

- Pan, X. *et al.* A DNA integrity network in the yeast *Saccharomyces cerevisiae*. *Cell* **124**, 1069–1081 (2006).
- Fiedler, D. *et al.* Functional organization of the *S. cerevisiae* phosphorylation network. *Cell* **136**, 952–963 (2009).
- Tong, A.H. *et al.* Systematic genetic analysis with ordered arrays of yeast deletion mutants. *Science* **294**, 2364–2368 (2001).
- Pan, X. *et al.* A robust toolkit for functional profiling of the yeast genome. *Mol. Cell* **16**, 487–496 (2004).
- Schuldiner, M. *et al.* Exploration of the function and organization of the yeast early secretory pathway through an epistatic miniarray profile. *Cell* **123**, 507–519 (2005).
- Roguev, A. *et al.* Conservation and rewiring of functional modules revealed by an epistasis map in fission yeast. *Science* **322**, 405–410 (2008).
- Typas, A. *et al.* High-throughput, quantitative analyses of genetic interactions in *E. coli*. *Nat. Methods* **5**, 781–787 (2008).
- Butland, G. *et al.* eSGA: *E. coli* synthetic genetic array analysis. *Nat. Methods* **5**, 789–795 (2008).
- Joshi, S.M. *et al.* Characterization of mycobacterial virulence genes through genetic interaction mapping. *Proc. Natl. Acad. Sci. USA* **103**, 11760–11765 (2006).
- Girgis, H. *et al.* A comprehensive genetic characterization of bacterial motility. *PLoS Genet.* **3**, 1644–1660 (2007).
- Lampe, D.J., Churchill, M. & Robertson, H. A purified mariner transposase is sufficient to mediate transposition *in vitro*. *EMBO J.* **15**, 5470–5479 (1996).
- Subramanian, A. *et al.* Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. *Proc. Natl. Acad. Sci. USA* **102**, 15545–15550 (2005).
- Jacobs, M.A. *et al.* Comprehensive transposon mutant library of *Pseudomonas aeruginosa*. *Proc. Natl. Acad. Sci. USA* **100**, 14339–14344 (2003).
- Fujita, Y. Carbon catabolite control of the metabolic network in *Bacillus subtilis*. *Biosci. Biotechnol. Biochem.* **73**, 245–259 (2009).
- Giammarinaro, P. & Paton, J. Role of RegM, a homologue of the catabolite repressor protein CcpA, in the virulence of *Streptococcus pneumoniae*. *Infect. Immun.* **70**, 5454–5461 (2002).
- Iyer, R., Baliga, N.S. & Camilli, A. Catabolite control protein A (CcpA) contributes to virulence and regulation of sugar metabolism in *Streptococcus pneumoniae*. *J. Bacteriol.* **187**, 8340–8349 (2005).
- Chapuy-Regaud, S. *et al.* RegR, a global LacI/GalR family regulator, modulates virulence and competence in *Streptococcus pneumoniae*. *Infect. Immun.* **71**, 2615–2625 (2003).
- Nieto, C., Espinosa, M. & Puyet, A. The maltose/maltodextrin regulon of *Streptococcus pneumoniae* differential promoter regulation by the transcriptional repressor MalR. *J. Biol. Chem.* **272**, 30860–30865 (1997).
- von Mering, C. *et al.* STRING 7—recent developments in the integration and prediction of protein interactions. *Nucleic Acids Res.* **35**, D358–D362 (2007).
- Mascher, T. *et al.* The *Streptococcus pneumoniae* cia regulon: CiaR target sites and transcription profile analysis. *J. Bacteriol.* **185**, 60–70 (2003).
- Throup, J.P. *et al.* A genomic analysis of two-component signal transduction in *Streptococcus pneumoniae*. *Mol. Microbiol.* **35**, 566–576 (2000).
- Sebert, M.E. *et al.* Microarray-based identification of *htrA*, a *Streptococcus pneumoniae* gene that is regulated by the CiaRH two-component system and contributes to nasopharyngeal colonization. *Infect. Immun.* **70**, 4059–4067 (2002).
- Drees, B.L. *et al.* Derivation of genetic interaction networks from quantitative phenotype data. *Genome Biol.* **6**, R38 (2005).
- St. Onge, R.P. *et al.* Systematic pathway analysis using high-resolution fitness profiling of combinatorial gene deletions. *Nat. Genet.* **39**, 199–206 (2007).
- King, S.J., Hippe, K.R. & Weiser, J. Deglycosylation of human glycoconjugates by the sequential activities of exoglycosidases expressed by *Streptococcus pneumoniae*. *Mol. Microbiol.* **59**, 961–974 (2006).
- Lampe, D.J. *et al.* Hyperactive transposase mutants of the Himar1 mariner transposon. *Proc. Natl. Acad. Sci. USA* **96**, 11428–11433 (1999).
- Bae, T. *et al.* Generating a collection of insertion mutations in the *Staphylococcus aureus* genome using bursa aurealis. *Methods Mol. Biol.* **416**, 103–116 (2008).
- Akerley, B.J. *et al.* Systematic identification of essential genes by *in vitro* mariner mutagenesis. *Proc. Natl. Acad. Sci. USA* **95**, 8927–8932 (1998).
- Sassetti, C.M., Boyd, D.H. & Rubin, E.J. Genes required for mycobacterial growth defined by high density mutagenesis. *Mol. Microbiol.* **48**, 77–84 (2003).

ONLINE METHODS

Bacterial strains, growth and media. The *S. pneumoniae* strain used in this study (AC2394) is an acapsular derivative of TIGR4, a serotype 4 clinical isolate. *S. pneumoniae* was grown in Todd Hewitt broth supplemented with yeast extract (THY) and 5 $\mu\text{l ml}^{-1}$ Oxyrase (Oxyrase, Inc) or on sheep's blood agar plates at 37 °C in a 5% CO₂ atmosphere. Where appropriate, blood plates contained 4 $\mu\text{g ml}^{-1}$ chloramphenicol (Cm) and/or 200 $\mu\text{g ml}^{-1}$ spectinomycin (Sp).

Construction of marked deletions. Gene knockouts of the five query genes *ccpA*, *regR*, *malR*, *malX* and SP_1683 were constructed by replacing the entire coding sequence with a Cm-resistance cassette. Single gene and double gene knockouts that we used to validate single gene fitness and genetic interactions with *ccpA* were constructed in the same manner as the query genes but instead an Sp-resistance cassette was used (see **Supplementary Table 6** for the strains used in this study). Construction of PCR products for gene replacement and transformation of *S. pneumoniae* were performed as described previously^{16,30}.

Transposon library construction. The *magellan5* mini-transposon is a derivative of the Himar1 Mariner transposon that contains a spectinomycin resistance marker. The mini-transposon is carried on a plasmid (pR412), which contains an ampicillin resistance marker and can be propagated in *E. coli*. A single nucleotide was changed in both inverted repeats of *magellan5* with the Quickchange Multi site-directed mutagenesis kit (Stratagene). The inverted repeat sequence was changed from 5'-agaccggggacttatcaGccaacctgt-3' to 5'-agaccggggacttatcaTccaacctgt-3' thereby introducing an MmeI restriction site (the mutation is in uppercase). The *magellan5* mini-transposon with the MmeI site is from now on referred to as *magellan6*.

Library construction was essentially performed as previously described³¹. Briefly, plasmid DNA harboring *magellan6* was purified from *E. coli* with the Qiagen mini plasmid preparation kit. *S. pneumoniae* DNA was isolated with the Qiagen Blood and Tissue kit from 3 ml of an exponentially growing culture. *In vitro* *magellan6* transposition reactions were carried out with purified MarC9 transposase, 1 μg of *S. pneumoniae* target DNA and 1 μg of *magellan6* plasmid DNA. Reactions were incubated for 1 h at 30 °C, inactivated for 20 min at 72 °C, ethanol-precipitated and resuspended in gap repair buffer (50 mM Tris (pH 7.8), 10 mM MgCl₂, 1 mM DTT, 100 nM dNTPs and 50 ng BSA). Transposition product gaps were repaired with *E. coli* DNA ligase overnight at 16 °C after which repaired transposition products were transformed into naturally competent *S. pneumoniae*. On the following day transformants were scraped off of blood plates and pooled into libraries of approximately 25,000 transformants per library for the wild-type strain and 10,000–25,000 transformants per library for the query strains (in total 6 libraries were screened for the wild type and three libraries for each query strain). To separate transformants from any residual agar, cells were grown in THY for a maximum of 2 h after which libraries were collected. From a fraction of the library, DNA was isolated (t_1) and another fraction was used to seed another culture in 10 ml of THY. Libraries were grown in THY for approximately seven generations to late exponential growth phase after which bacteria were collected and DNA was isolated (t_2). The exact expansion of each library from

t_1 to t_2 , was determined by plating dilutions of the culture from both time points on blood agar plates. Note that both the transformation procedure and the 2 h growth to separate transformants from residual agar may place a limit on the dynamic range of mutants that we were able to identify by Tn-seq. Specifically, mutants with fitness values less than 0.5 were not identified in our study presumably owing to their outcompetition during these early steps. It is likely that changes to the protocol that reduce the number of generations before selection at t_1 , will increase the dynamic range.

Sample preparation and Illumina sequencing. DNA from two time points, before selection (t_1) and after selection (t_2), was digested for 2.5 h at 37 °C with MmeI (NEB). To prevent re-ligation in the subsequent step, the 5' phosphate group was removed with Calf intestinal alkaline phosphatase (NEB) after which DNA was phenol-chloroform-extracted, ethanol-precipitated and dissolved in H₂O. Next, an adaptor was ligated to the overhang left by MmeI overnight at 16 °C after which DNA was again phenol-chloroform-extracted, ethanol-precipitated and dissolved in H₂O. A PCR was performed with the adaptor-ligated samples as template and with primers of which one is complementary to the adaptor and the other to the mini-transposon inverted repeat sequence. Both primers also contained Illumina-specific sequences that have complementarity to two specific primers in the Illumina flow cell and enable eventual sequencing. Although it is essential that the adaptor sequence contains one of the Illumina-specific sequences, the rest of the adaptor can vary. Therefore it is possible to include a barcode sequence in the adaptor that enables mixing of samples in a single flow cell lane and subsequent separation after sequencing. The resulting PCR product was 160 bp in length and was amplified using the following parameters: 95 °C for 30 s, followed by 22 cycles of 10 s at 95 °C, 25 s at 55 °C and 15 s at 72 °C, followed by 10 min at 72 °C (through Illumina sequencing of PCR products amplified with 19, 22 and 25 cycles we determined that the PCR remained linear at least up to 25 cycles). The PCR product was gel purified, dissolved in H₂O and sequenced on an Illumina Genome Analyzer II according to the manufacturers protocol (Illumina). One to three libraries from a single time point were sequenced in a single flow cell lane. After 30 sequencing cycles, raw data was extracted, split into different samples based on the 6-nucleotide-barcode sequence and stripped from the barcode and four nucleotides of the adaptor sequence. This resulted in 6–10 $\times 10^6$, 20-base *S. pneumoniae*-specific reads per flow cell lane.

Fitness calculation. After sequencing, 20-bp reads were mapped to the *S. pneumoniae* TIGR4 genome with the program Bowtie³². Bowtie parameters were set such that reads could contain a single mismatch but were only allowed if they mapped to a unique location. If mapping to multiple sites was possible, the read was excluded from the analyses. In addition, sequences from the first two wild-type libraries were also analyzed with the program CLC genomics workbench v2.1.1 (CLC bio) and the program MAQ³³, which confirmed the Bowtie analyses. In addition, sequences from all libraries were visually checked with the program CLC genomics workbench. Owing to the short read length, the error rate was low, and on average only 5% of the reads had to be discarded for two reasons: (i) 4.5% of these 'reads' could not be mapped

to a single location (these reads mapped to 119 endogenous transposon-related genes and 11 other repeated sequences (see **Supplementary Table 7** for removed genes)), and (ii) the remaining reads (0.5%) did not map anywhere and were categorized as junk sequences. For the two time points (t_1 and t_2) the number of reads at each transposon insertion location were recorded; on average 82 reads were mapped per insertion at each time point. As an insertion with a low number of reads that slightly fluctuates over time could influence the data disproportionately, only insertions with eight or more reads at one time point were included in the analyses. This resulted in excluding 2.5% of the insertions. Lowering the limit to a minimum of five reads or setting the limit higher to a minimum of 15 reads did not considerably alter the results, which suggests that 8 is a robust limit. Subsequently the data were normalized by equalizing the total number of sequenced reads per time point. The number of reads per time point never varied more than 10% and therefore the normalization factors for all datasets were small (1.03–1.10). The change over time in the number of reads at a specific location was then used to calculate fitness. Thus for each insertion, fitness W_i was calculated by comparing the fold expansion of the mutant relative to the rest of the population with the following equation^{34,35}:

$$W_i = \frac{\ln(N_i(t_2) \times d / N_i(t_1))}{\ln((1 - N_i(t_2)) \times d / (1 - N_i(t_1)))}$$

in which $N_i(t_1)$ and $N_i(t_2)$ are the frequency of the mutant in the population at the start and at the end of the experiment, respectively, and d (expansion factor) represents the growth of the bacterial population during library selection. The expansion factor was directly measured for each library by plating appropriate dilutions of the bacterial library at t_1 and t_2 and dividing the number of bacteria counted at each time point ($d = \text{number of bacteria at } t_2 / \text{number of bacteria at } t_1$).

After fitness was determined for each insertion, data were normalized against a set of genes that are inactive in *S. pneumoniae* and have no fitness effect (**Supplementary Table 2**). These genes are degenerate transposon-related sequences and disrupting them with a transposon will not result in a fitness effect. As the fitness of these inactive genes is therefore equal to that of the wild-type background strain in which the transposon library is generated, they can be used as a reference point to which the rest of the library is normalized. To achieve this, fitness is calculated over all these genes and their average fitness is subsequently set to 1. The same factor is used to normalize the rest of the dataset, thus setting all fitness values relative to 1. The normalization factors used for all datasets were small (0.981–1.014). This shows that fitness for the reference genes was already close to 1, which underscores the fact that these genes are neutral. Note that in cases in which a query strain was used as the background strain, the average of the reference genes was set to the fitness of the query strain (*ccpA* fitness = 0.84, *regR* fitness = 1.0, *SP_1683* fitness = 0.98, *malX* fitness = 1.04 and *malR* fitness = 1.05). Finally all of the insertions in a gene were used to calculate a gene's average fitness and s.d. To further control for deviations in fitness owing to insertions with small numbers of reads, a weighted average was used. This meant that insertions with 50 or less reads received a proportional lower weight. This slightly increased correlations between replicates and it slightly lowered s.d. A generally used method to

determine fitness is the competition index, in which the change in mutant to wild type ratio over time is determined^{36,37}. The advantage of using the equation above and including the expansion factor is that W_i now represents the actual growth rate per generation, which makes fitness independent of time, more accurate and enables comparisons between experiments.

To determine whether fitness effects significantly differed from wild type and were advantageous or disadvantageous, three requirements had to be fulfilled: (i) fitness was calculated from at least 10 transposon insertions (genes with less than 10 insertions were excluded), (ii) fitness had to deviate from wild-type strain fitness (value of 1) by at least 4%, and (iii) fitness had to be significantly different from wild type in a one sample *t*-test with Bonferroni correction. All significant deviations therefore had a *P* value < 0.00003 (*P* value / number of comparisons). Genes were listed as “possibly essential” if no insertions were recorded or less than three insertions were recorded, and the gene size was ≥ 400 bp.

1 × 1 competition assays and single strain growth. For 1 × 1 competitions bacteria were recovered from blood agar, sub-cultured in THY and grown at 37 °C for 1.5 h to an OD₆₀₀ of 0.1–0.2. Subsequently two strains were mixed in a 1:1 ratio. Mutants were always competed against their background strain; transposon insertions and strains with a single gene drug marker replacement were competed against the wild-type strain, and double mutants (which all had *ccpA* deleted) were competed against the *ccpA* mutant. Competitions were grown for approximately 7 generations to late exponential growth phase, and ratios of the competing strains were determined at the start and at the end of the competition by plating appropriate dilutions on blood agar plates with selective antibiotics. Competitions were performed no less than 5 times.

Single strain growth was performed in THY no less than four times. For each strain, OD₆₀₀ measurements were taken every half hour and were used to calculate doubling times for 10 mutant strains and the wild type.

Genetic interactions. Genetic interactions are defined as a deviation from the multiplicative model, which states that if a strain deleted for gene *i* has a fitness W_i and a strain deleted for gene *j* has a fitness W_j , then the double mutant strain W_{ij} is expected to have the fitness $W_i \times W_j$ (note that at least one strain's fitness (W_i or W_j) needs to deviate significantly from 1). Because we wanted to minimize the number of false positives, genetic interactions had to fulfill three requirements: (i) fitness was calculated from at least 10 transposon insertions (genes with less than 10 insertions were excluded), (ii) expected fitness and observed fitness had to deviate by at least 10% and (iii) significant interactions had to pass the Student's *t*-test with Bonferroni correction for multiple testing, which approximately lowers the level of a significant *P* value to 0.05/*n* (*n* = the number of comparisons). All significant interactions therefore had a *P* value < 0.00003. Significant genetic interactions could be further split into seven categories^{23,24}: synergistic ($W_i = W_j < W_{ij}$), partial masking ($W_i < W_j < W_{ij}$), masking ($W_i < W_j = W_{ij}$), partial suppression ($W_i < W_{ij} < W_j$), suppression ($W_i = W_{ij} < W_j$), antagonistic ($W_{ij} < W_i \leq W_j$) and co-equal ($W_i = W_j = W_{ij}$). Interactions were visualized with the program Cytoscape³⁸ and

assigned to a category based on a Student's t -test with Bonferroni correction for multiple testing. In addition, arrows in the networks depict the hypothesized direction of the interaction; for the categories partial masking and masking, W_j was the source and W_i was the target, which means that the gene with the highest fitness (W_j) was masking the effect of the gene with the lower fitness (W_i). For the categories partial suppression and suppression, W_i was the source and W_j was the target; here the gene with the lowest fitness (W_i) was suppressing the effect of the gene with the highest fitness (W_j). The source was thus the node where the arrow originates and the target where the arrow ends.

30. Bricker, A.L. & Camilli, A. Transformation of a type 4 encapsulated strain of *Streptococcus pneumoniae*. *FEMS Microbiol. Lett.* **172**, 131–135 (1999).
31. Hava, D.L. & Camilli, A. Large-scale identification of serotype 4 *Streptococcus pneumoniae* virulence factors. *Mol. Microbiol.* **45**, 1389–1406 (2002).
32. Langmead, B. *et al.* Ultrafast and memory-efficient alignment of short DNA sequences to the human genome. *Genome Biol.* **10**, R25 (2009).
33. Li, H., Ruan, J. & Durbin, R. Mapping short DNA sequencing reads and calling variants using mapping quality scores. *Genome Res.* **18**, 1851–1858 (2008).
34. Lenski, R.E., Rose, M.R.S. & Tadler, S.C. Long-term experimental evolution in *Escherichia coli*. I. Adaptation and divergence during 2,000 generations. *Am. Nat.* **138**, 1315–1341 (1991).
35. van Opijnen, T., Boerlijst, M.C. & Berkhout, B. Effects of random mutations in the human immunodeficiency virus type 1 transcriptional promoter on viral fitness in different host cell environments. *J. Virol.* **80**, 6678–6685 (2006).
36. Hava, D.L. & Camilli, A. Large-scale identification of serotype 4 *Streptococcus pneumoniae* virulence factors. *Mol. Microbiol.* **45**, 1389–1406 (2002).
37. Marcusson, L.L. *et al.* Interplay in the selection of fluoroquinolone resistance and bacterial fitness. *PLoS Pathog.* **5**, e1000541 (2009).
38. Shannon, P. Cytoscape: a software environment for integrated models of biomolecular interaction networks. *Genome Res.* **13**, 2498–2504 (2003).